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Endoscope camera head

Abstract

An endoscope camera head **6** includes a first and a second members **8**, **9** that are relatively rotatable about a first axis Ax**1**. The first member **8** has a first rotation-sliding surface **84** that extends in an annular shape about the first axis Ax**1** as its center. The second member **9** has a second rotation-sliding surface **93** that engages with the first rotation-sliding surface **84** in a state in which movement to a direction of the first axis Ax**1** relative to the first member **8** is restricted, and that rotation-slides relative to the first rotation-sliding surface **84** about the first axis Ax**1**. The second rotation-sliding surface **93** abuts only on a part of the first rotation-sliding surface **84**, to expose a part of the first rotation-sliding surface **84** outside.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. application Ser. No. 16/633,587, filed Jan. 24, 2020, which is based on PCT filing PCT/JP2018/018989, filed May 16, 2018, and claims priority to Japanese Patent Application No. 2017-148486, filed Jul. 31, 2017, the entire contents of each are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention relates to an endoscope camera head that is used in an endoscope device to observe an inside of a subject, such as a human body and a mechanical structure.

BACKGROUND ART

(2) Endoscope devices to observe inside a subject, such as a human body and a mechanical structure in a medical field or an industrial field has conventionally been known (for example, refer to Patent Literature 1).

(3) An endoscope device described in Patent Literature 1 includes an endoscope (optical telescope) that captures and emits a subject image inside the subject, and an endoscope camera head (TV camera) that holds the endoscope and that forms an image from the subject image emitted from the endoscope. Moreover, the endoscope is rotatably held about an optical axis set inside with respect to the endoscope camera head.

(4) Specifically, the endoscope camera head includes a coupler and a camera-head main unit.

(5) The coupler has a bottomed-cylindrical shape in which an eyepiece of the endoscope can be engaged. Furthermore, at a bottom portion of the coupler, a through hole in a circular shape when viewed from top, piercing therethrough to both sides is formed.

(6) The camera-head main unit includes a casing, and an imaging unit that is housed in the casing, and that forms an image of the subject image emitted from the endoscope. On an external surface of the casing, a concave portion that extends in a ring shape about a first axis as a center is formed.

(7) Furthermore, the coupler (endoscope) is configured such that a rim portion of the through hole in the coupler engages with the concave portion of the casing, and that an outer surface of the rim portion slides on an inner surface (hereinafter, referred to as coupler rotation-sliding surface) of the concave portion, thereby rotating with respect to the camera-head main unit about the first axis as

the center.

CITATION LIST

Patent Literature

(8) Patent Literature 1: JP-A-2000-227559

DISCLOSURE OF INVENTION

Technical Problem

(9) However, in the endoscope camera head described in Patent Literature 1, the coupler rotation-sliding surface is concealed with the casing rotation-sliding surface, and is not exposed outside. Therefore, when cleaning the endoscope camera head, it is difficult to clean a portion between the coupler rotation-sliding surface and the casing rotation-sliding surface with a brush. Moreover, it is difficult to let cleaning solution enter between the coupler rotation-sliding surface and the casing rotation-sliding surface. Accordingly, there is a problem that much time is consumed for cleaning.

(10) The present invention is achieved in view of the above problem, and it is an object of the present invention to provide an endoscope camera head that enables to reduce cleaning time.

Solution to Problem

(11) To solve the above-described problem and achieve the object, an endoscope camera head according to the present invention includes a first member and a second member that are capable of rotating relative to each other about a first axis, wherein the first member has a first rotation-sliding surface that extends in an annular shape about the first axis as a center, the second member engages with the first rotation-sliding surface in a state in which movement to a direction of the first axis is restricted relative to the first member, and has a second-rotation sliding surface that rotation-slides relatively to the first rotation-sliding surface about the first axis, and the second rotation-sliding surface abuts only on a part of the first rotation-sliding surface, to expose a part of the first rotation-sliding surface outside.

(12) Moreover, in the above-described endoscope camera head according to the present invention, in the first member, a concave portion that extends in an annular shape about the first axis as a center is provided, the first rotation-sliding surface includes an inner surface of the concave portion, in the second member, a convex portion that extends in an annular shape about the first axis as a center, and that engages with the concave portion is provided, the second rotation-sliding surface includes an outer surface of the convex portion, and at a distal end of the convex portion, a groove portion that extends in a spiral shape about the first axis as a center, and respective parts of both end portions of which in a direction of the first axis are respectively positioned outside the concave portion is formed.

(13) Moreover, in the above-described endoscope camera head according to the present invention, in the first member, a concave portion that extends in an annular shape about the first axis as a center is provided, the first rotation-sliding surface includes an inner surface of the concave portion, in the second member, a projection portion, a distal end of which extends in an arc shape about the first axis as a center, and that engages with the concave portion is provided, and the second rotation-sliding surface includes an outer surface of the projection portion.

(14) Moreover, in the above-described endoscope camera head according to the present invention, the projection portion is provided in plurality in a direction of circle about the first axis as a center.

(15) Moreover, the above-described endoscope camera head according to the present invention includes: a coupler holding an endoscope that captures and outputs a subject image; and a camera-head main unit that includes a casing supporting the coupler rotatably about the first axis, and an imaging unit that is housed in the casing, and that images the subject image emitted from the endoscope, wherein the first member is the casing, and the second member is the coupler.

(16) Moreover, the above-described endoscope camera head according to the present invention includes: a coupler holding an endoscope that captures and outputs a subject image; and a camera-head main unit that includes a casing supporting the coupler rotatably about the first axis, and an imaging unit that is housed in the casing, and that images the subject image emitted from the

endoscope, wherein the first member is the coupler, and the second member is the casing.

Advantageous Effects of Invention

(17) In the endoscope camera head according to the present invention, the first and the second members constituted of a coupler and a casing, respectively have the first and the second rotation sliding surfaces. The second rotation-sliding surface abuts only on a part of the first rotation-sliding surface, to expose a part of the first rotation-sliding surface outside.

(18) Therefore, it is possible to insert a cleaning brush or to let cleaning solution reach the second rotation-sliding surface through the part of the first rotation-sliding surface exposed outside.

Furthermore, by rotating the first and the second members relatively to each other about the first axis in a state in which the cleaning brush is inserted or the cleaning solution has entered therein, the first and the second rotation-sliding surfaces can be easily and efficiently cleaned.

(19) Therefore, according to the endoscope camera head according to the present invention, an effect of reducing cleaning time is produced.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram illustrating a schematic configuration of an endoscope device according to a first embodiment.

(2) FIG. 2 is a cross-section illustrating a connecting portion between an eyepiece and an endoscope camera head.

(3) FIG. 3 is an enlarged view of a part of FIG. 2.

(4) FIG. 4 is a diagram illustrating a groove portion.

(5) FIG. 5 is a diagram illustrating a modification 1-1 of the first embodiment.

(6) FIG. 6 is a diagram illustrating the modification 1-1 of the first embodiment.

(7) FIG. 7 is a cross-section illustrating a connecting portion between an eyepiece and an endoscope camera head according to a second embodiment.

(8) FIG. 8 is a diagram illustrating a coupler according to the second embodiment.

(9) FIG. 9 is a diagram illustrating an effect of the second embodiment.

(10) FIG. 10 is a diagram illustrating an effect of the second embodiment.

(11) FIG. 11 is a diagram illustrating a modification 2-1 of the second embodiment.

(12) FIG. 12 is a diagram illustrating the modification 2-1 of the second embodiment.

(13) FIG. 13 is a diagram illustrating a modification 2-2 of the second embodiment.

(14) FIG. 14 is a diagram illustrating a modification 2-3 of the second embodiment.

(15) FIG. 15 is a diagram illustrating a modification 3-1 of the first and the second embodiments.

BEST MODE FOR CARRYING OUT THE INVENTION

(16) Hereinafter, modes (hereinafter, embodiments) to implement the present invention will be described with reference to the drawings. Note that embodiments described below are not intended to limit the present invention. Furthermore, like reference symbols are assigned to like parts in descriptions of the drawings.

First Embodiment

(17) Schematic Configuration of Endoscope Device FIG. 1 is a diagram illustrating a configuration of an endoscope device 1 according to a first embodiment.

(18) The endoscope device 1 is used in a medical field, and is a device to treat a living tissue (incision, and the like) while observing inside the living body. This endoscope device 1 includes, as illustrated in FIG. 1, a resectoscope 2, an endoscope imaging device 3, a display device 4, and a control device 5.

(19) The resectoscope 2 is a part that is inserted into a living body, and that captures a subject image and treats a living tissue. This resectoscope 2 includes, as illustrated in FIG. 1, a sheath 21, a guide

tube 22, an endoscope 23, a resect electrode member 24, and a handle portion 25.

(20) The sheath 21 is a part having a cylindrical shape, and inserted into a living body.

(21) The guide tube 22 has a smaller outer diameter than an inner diameter of the sheath 21, and is inserted in the sheath 21. The guide tube 22 is fixed to the sheath 21 through a mounting member 221 (FIG. 1) on its distal end side (left side in FIG. 1).

(22) In the mounting member 221, a water supply inlet 222 to infuse solution into the sheath 21, and to supply the solution from a distal end of the sheath 21 is provided.

(23) The endoscope 23 is a part of capturing a subject image, and includes, as illustrated in FIG. 1, an insertion portion 231 and an eyepiece 232.

(24) The insertion portion 231 is fixed inside the guide tube 22, and is inserted into the sheath 21. Inside this insertion portion 231, an optical system that is constituted of one or more lenses, and that gathers the subject image is provided.

(25) The eyepiece 232 is connected to a proximal end (right end portion in FIG. 1) of the insertion portion 231. Inside this eyepiece 232, an eyepiece optical system 2321 (refer to FIG. 2) that emits the subject image gathered by the optical system inside the insertion portion 231 from the eyepiece 232 to outside is provided. The eyepiece 232 is formed in a tapered shape in which a diameter increases toward a right side, and to an increased-diameter portion, the endoscope imaging device 3 is detachably connected.

(26) To the eyepiece 232, a light source connector 2323 to connect a light guide 2322 is arranged. That is, light supplied to the light guide 2322 from a light source device (not shown) is supplied to the insertion portion 231 through the eyepiece 232. The light supplied to the insertion portion 231 is emitted from the distal end of the insertion portion 231, and is irradiated inside a living body. The light irradiated inside the living body and reflected in the living body (subject image) is emitted from the eyepiece 232 through the optical system inside the insertion portion 231 and the eyepiece optical system 2321.

(27) The resect electrode member 24 is inserted in the sheath 21 through the mounting member 221, and its distal end protrudes out from the distal end of the sheath 21. The resect electrode member 24 brought in contact with a living tissue at its end portion, and treats the living tissue with a high frequency current.

(28) The handle portion 25 is a part with which a doctor or the like holds the resectoscope 2, and operates the resect electrode member 24. This handle portion 25 includes, as illustrated in FIG. 1, a fixing ring 251, a slider 252, and a spring member 253.

(29) The fixing ring 251 is a part on which a doctor or the like hooks his/her thumb, and is fixed to the guide tube 22.

(30) The slider 252 has the guide tube 22 inserted therethrough, to be configured to be movable in a left and right direction in FIG. 1 along the guide tube 22.

(31) To this slider 252, the resect electrode member 24 is fixed as illustrated in FIG. 1. That is, the resect electrode member 24 reciprocates in a left and right direction in FIG. 1 inside the sheath 21, along with movement of the slider 252.

(32) Moreover, in the slider 252, a power source connector 2522 to connect a high-frequency power cord 2521 connected to a high frequency power source (not shown) is provided. This power source connector 2522 is electrically connected with the resect electrode member 24 through a lead (not shown).

(33) Furthermore, in the slider 252, a finger hook member 2523 on which fingers other than a thumb of a doctor or the like are hooked, to move the slider 252 (to reciprocate the resect electrode member 24) is arranged.

(34) The spring member 253 has a substantially U-shape, and one end there of is attached to the fixing ring 251, and the other end is attached to the slider 252. The spring member 253 applies pressure in such a direction that the slider 252 is apart from the fixing ring 251.

(35) That is, a doctor or the like hooks his/her fingers on the fixing ring 251 and the finger hook

member **2523**, and pulls the finger hook member **2523**, resisting the pressure of the spring member **253**, to thereby move the slider **252** in rightward in FIG. **1** (to move the resect electrode member **24** rightward in FIG. **1**). On the other hand, when the doctor or the like moves the fingers off the finger hook member **2523**, the slider **252** (the resect electrode member **24**) moves leftward in FIG. **1** by the pressure of the spring member **253**.

(36) The endoscope imaging device **3** is detachably connected to the eyepiece **232** of the resectoscope **2** (the endoscope **23**). The endoscope imaging device **3** images a subject image (a subject image emitted from the eyepiece **232**) captured by the endoscope **23**, and outputs an image signal (RAW signal) from this imaging under control of the control device **5**. The image signal is, for example, an image signal of **4K** or higher.

(37) A detailed configuration of the endoscope imaging device **3** will be described later.

(38) The display device **4** is constituted of a display using a liquid crystal, an organic EL (electroluminescence), or the like, and displays, under a control of the control unit **5**, an observation image based on a video signal from the control unit **5**.

(39) The control device **5** includes a (CPU) central processing unit, and controls operations of the endoscope imaging device **3**, the display device **4**, and the light source device (not shown) in a centralized manner. For example, the control device **5** generates a video signal for display by subjecting the image signal (RAW signal) output from the endoscope imaging device **3** to predetermined image processing. The control device **5** then causes the display device **4** to display an observation image based on the video signal.

(40) Configuration of Endoscope Imaging Device

(41) Next, a configuration of the endoscope imaging device will be described.

(42) FIG. **2** is a cross-section illustrating a connecting portion between the eyepiece **232** and an endoscope camera head **6**. FIG. **3** is an enlarged view of a part of FIG. **2**. The endoscope imaging device **3** includes, as illustrated in FIG. **1** or FIG. **2**, the endoscope camera head **6** and a cable **7** (FIG. **1**).

(43) The endoscope camera head **6** is a part detachably connected to the eyepiece **232** as illustrated in FIG. **1** or FIG. **2**. This endoscope camera head **6** includes, as illustrated in FIG. **2**, a casing **8**, a coupler **9**, a prism **10**, a lens unit **11**, and an imaging unit **12**.

(44) The casing **8** corresponds to a first member according to the present invention. This casing **8** includes, as illustrated in FIG. **2** or FIG. **3**, a casing main body **81** and an attachment bush **82**.

(45) The casing main body **81** is a case that houses the respective parts **10** to **12**. In this casing main body **81**, a protrusion portion **811** in a cylindrical shape, communicating inside and outside of the casing main body **81** is arranged, as illustrated in FIG. **2** or FIG. **3**.

(46) In the protrusion portion **811**, as illustrated in FIG. **2** or FIG. **3**, a jut-out portion **812** that juts out toward a left side in FIG. **2** and FIG. **3** is arranged on an outer circumferential side. This jut-out portion **812** has an outer diameter substantially the same as the protrusion portion **811**, is formed in a cylindrical shape having an inner diameter larger than the protrusion portion **811**, and is integrated with a distal end of the protrusion portion **811**, coaxially with the protrusion portion **811**. Moreover, on an inner circumferential surface of the jut-out portion **812**, a screw groove **813** is formed.

(47) The attachment bush **82** is a part to attach the coupler **9** to the casing **8**. This attachment bush **82** includes, as illustrated in FIG. **2** or FIG. **3**, a bush main body **821** and a jut-out portion **822**.

(48) The bush main body **821** has an inner diameter substantially the same as the protrusion portion **811**, and is formed in a cylindrical shape having an outer diameter substantially the same as the inner diameter of the jut-out portion **812**. Furthermore, on an outer circumferential surface of the bush main body **821**, a screw groove **823** is formed as illustrated in FIG. **2** or FIG. **3**.

(49) The jut-out portion **822** extends out from a rim portion at a left end in FIG. **2**, FIG. **3** on the outer circumferential surface of the bush main body **821**, and is formed in an annular shape having an outer diameter substantially the same as the jut-out portion **812**.

(50) The casing main body **81** and the attachment bush **82** are fixed to each other, by engaging the screw grooves **813** and **823** with each other. In this state, center axes of the casing main body **81** and the attachment bush **82** agree with each other. The center axis corresponds to a first axis Ax1 (FIG. 2, FIG. 3) according to the present invention. Moreover, in the attachment bush **82**, an optical device **83** of sapphire glass, or the like is fixed as illustrated in FIG. 2 or FIG. 3.

(51) In the state in which the casing main body **81** and the attachment bush **82** are fixed to each other, with surfaces of respective ends of the jut-out portion **812** and **822** facing each other, and the outer circumferential surface of the bush main body **821**, a concave portion **84** (FIG. 2, FIG. 3) extending in an annular shape having the first axis Ax1 in center is provided. An inner surface of the concave portion **84** (the surfaces of respective ends of the jut-out portions **812** and **822** facing each other, and the outer circumferential surface of the bush main body **821**) corresponds to a first rotation-sliding surface according to the present invention.

(52) The coupler **9** is a part corresponding to a second member according to the present invention. This coupler **9** has a bottomed-cylindrical shape in which the eyepiece **232** can be fitted therein as illustrated in FIG. 2.

(53) On an inner circumferential surface of this coupler **9**, a pressurizing portion **91** is arranged as illustrate in FIG. 2.

(54) This pressurizing portion **91** has elasticity enable to move in a direction of separation in adjoining isolation relative to a center axis Ax2 of the coupler **9**, and applies pressure to the eyepiece **232** toward a bottom portion of the coupler **9**, by abutting on an outer circumferential surface of the eyepiece **232** that is fitted in the coupler **9**. In the first embodiment, four pieces of the pressurizing portions **91** are provided, and are arranged to be rotationally symmetric at 90° about the center axis Ax2. Note that in the state in which the eyepiece **232** is fitted inside the coupler **9**, an optical axis Ax3 of the endoscope **23** (FIG. 2, hereinafter, referred to as endoscope optical axis Ax3) agree with the center axis Ax2. Moreover, on an outer circumferential surface of the eyepiece **232** on a right end portion in FIG. 2, an expanding portion **2324** that expands toward a right side in FIG. 2 is provided on the entire circumference. Therefore, when the eyepiece **232** is pressurized toward the bottom portion of the coupler **9** by the pressurizing portion **91**, the expanding portion **2324** abuts on the bottom portion. Thus, space Ar (FIG. 2) is formed between the optical device **83** and the eyepiece optical system **2321**.

(55) Moreover, the bottom portion of the coupler **9** is constituted of a plate in a circular shape when viewed from top, each plane of which is perpendicular to the center axis Ax2. Moreover, the bottom portion has a thickness substantially the same as a width of the concave portion **84**. Furthermore, in the bottom portion, a through hole **92** that penetrates through to its front and rear surfaces, and that has a circular shape when viewed from top having an inner diameter substantially the same as an outer diameter of the bush main body **821** is formed. The coupler **9** is attached to the casing **8** as a rim portion of the through hole **92** engages with (fits in) the concave portion **84** in a state in which an opening of the bottomed-cylindrical shape is directed to the left side in FIG. 2. In this state, the center axis Ax2 agrees with the first axis Ax1. Moreover, the coupler **9** is restricted its movement to a direction of the first axis Ax1 relative to the casing **8**, and is rotatable relative to the casing **8** about the first axis Ax1 (the center axis Ax2).

(56) Therefore, the endoscope camera head **6** is structured rotatable about the first axis Ax1 (the center axis Ax2, the endoscope optical axis Ax3), relative to the eyepiece **232** of the endoscope **23** through the coupler **9**. Moreover, the endoscope camera head **6** is structured such that its center of gravity O (FIG. 1) is positioned off the first axis Ax1 (a rotation center axis relative to the eyepiece **232**). The endoscope camera head **6** rotates about the first axis Ax1 irrespective of rotation about the first axis Ax1 of the resectoscope **2**, and is structured to maintain an orientation in which an optical axis Ax4 (FIG. 2, hereinafter, referred to as in-case optical axis Ax4) set inside the casing **81** is along a vertical direction (orientation in which the center of gravity O is positioned below the first axis Ax1) all the time.

(57) The rim portion of the through hole **92** extends in an annular shape about the first axis Ax1, and corresponds to a convex portion **93** (FIG. 2, FIG. 3) according to the present invention. Furthermore, the outer surface of the convex portion **93** is a portion that slides on the inner surface of the concave portion **84**, and corresponds to a second rotation-sliding surface according to the present invention.

(58) At a distal end of the convex portion **93** (inner surface of the through hole **92**), a groove portion **94** is formed as illustrated in FIG. 2 or FIG. 3.

(59) FIG. 4 is a diagram illustrating the groove portion **94**. The groove portion **94** extends in a spiral shape about the first axis Ax1 (the center axis Ax2) at the distal end of the convex portion **93** as illustrated in FIG. 4. In the state in which the coupler **9** is attached to the casing **8**, a part of both end portions **941**, **942** in the direction of the first axis Ax1 in the groove portion **94** is respectively positioned outside the concave portion **84**.

(60) That is, in the state in which relative rotation of the casing **8** and the coupler **9** about the first axis Ax1 is stopped, the outer surface of the convex portion **93** (the second rotation-sliding surface according to the second embodiment) abuts only on a part of the inner surface (the first rotation-sliding surface according to the present invention) of the concave portion **84** by having the groove portion **94**, enabling to expose a part of the inner surface of the concave portion **84** outside.

(61) The prism **10** is arranged on the first axis Ax1 and on the in-case optical axis Ax4 as illustrated in FIG. 2, and polarizes a subject image captured by the endoscope **23** to change the traveling direction. Specifically, the prism **10** polarizes a subject image (subject image that travels along the endoscope optical axis Ax3) that is emitted from the eyepiece **232** and taken into the casing **8** through the optical device **83**, to change the traveling direction by 90° to travel along the in-case optical axis Ax4.

(62) The lens unit **11** is arranged on the in-case optical axis Ax4 as illustrated in FIG. 2. This lens unit **11** is constituted of one or plural lenses, and forms an image of the subject image entering through the prism **10** on an imaging surface of the imaging unit **12**. Moreover, the lens unit **11** is provided with an optical zoom function (not shown) that changes an angle of view by moving the one or plural lenses under control of the control device **5** or an operating unit **13** (FIG. 1), and a focus mechanism (not shown) that changes a focal point.

(63) The imaging unit **12** is arranged on the in-case optical axis Ax4 as illustrated in FIG. 2. The imaging unit **12** images a subject image formed by the lens unit **11** under control of the control device **5**. This imaging unit **12** is constituted of a sensor chip in which an imaging device (not shown) that receives a subject image formed by the lens unit **11** to convert into an electrical signal, such as a CCD (charge coupled device) and a CMOS (complementary metal oxide semiconductor), and a signal processing unit (not shown) that outputs an image signal by subjecting the electrical signal (analog signal) from the imaging device to a signal processing (A/D conversion, and the like), and the like are integrally formed, and outputs the image signal (RAW signal (digital signal)) subjected to the A/D conversion. Note that the signal processing unit described above may be prepared as a separate unit without integrating with the imaging device.

(64) The cable **7** is detachably connected at its one end to the control device **5** through a connector CN1 (FIG. 1), and is detachably connected at the other end to the endoscope camera head **6** through a connector CN2 (FIG. 1). The cable **7** transmits an image signal output from the endoscope camera head **6** to the control device **5**, and transmits a control signal, a synchronization signal, a clock, power, and the like output from the control device **5** to the endoscope camera head **6**.

(65) Note that in transmission of an image signal from the endoscope camera head **6** through the cable **7**, the image signal may be transmitted in an optical signal, or may be transmitted in an electrical signal. The same applies to transmission of the control signal, the synchronization signal, and the clock from the control device **5** to the endoscope camera head **6** through the cable **7**.

(66) Moreover, in the cable **7**, the operating unit **13** that receives various kinds of operations from a

doctor and the like (for example, an instruction of image quality adjustment (white balance adjustment, brightness adjustment, and the like) of an observation image, an instruction to change an angle of view or a focal point of the lens unit **11**, and the like) is provided as illustrated in FIG. **1**.

(67) According to the first embodiment described above, following effects are produced.

(68) The endoscope camera head **6** according to the first embodiment includes the casing **8** having the concave portion **84** that extends in an annular shape about the first axis Ax**1**, and the coupler **9** having the convex portion **93** that extends in an annular shape about the first axis Ax**1** and that engages with the concave portion **84**. Moreover, at the distal end of the convex portion **93**, the groove portion **94** is formed that extends in a spiral shape about the first axis Ax**1**, and in which respective parts of the both end portions **941** and **942** in the direction of the first axis Ax**1** are positioned outside the concave portion **84**, respectively. The outer surface of the convex portion **93** abuts only on a part of the inner surface of the concave portion **84** by having the groove portion **94**, in the state in which relative rotation of casing **8** and the coupler about the first axis Ax**1** is stopped, to expose a part of the inner surface of the concave portion **84**.

(69) Therefore, by relatively rotating the casing **8** and the coupler **9** about the first axis Ax**1** in a state in which the endoscope camera head **6** is immersed in cleaning solution, the cleaning solution flows into the groove portion **94** from the end portion **941** as indicated by an arrow in FIG. **3**, and follows the groove portion **94** to flows out of the groove portion **94** from the end portion **942**. That is, it is possible to clean a portion between the inner surface of the concave portion **84** and the outer surface of the convex portion **93** easily and effectively. Note that depending on a rotation direction of the casing **8** and the coupler **9**, a flowing direction of the cleaning solution indicated by the arrow in FIG. **3** is to be opposite.

(70) Therefore, according to the endoscope camera head **6** according to the first embodiment, an effect of reducing the cleaning time is produced.

Modification 1-1 of First Embodiment

(71) FIG. **5** and FIG. **6** are diagrams illustrating a modification 1-1 of the first embodiment.

Specifically, FIG. **5** is a diagram corresponding to FIG. **3**. FIG. **6** is a diagram illustrating a groove portion **815** according to the modification 1-1.

(72) In the endoscope camera head **6** according to the first embodiment described above, the first member according to the present invention is the casing **8**, and the second member according to the present invention is the coupler **9**, but it is not limited thereto. For example, as an endoscope camera head **6A** according to the modification 1-1 illustrated in FIG. **5** and FIG. **6**, the first member according to the present invention may be a coupler **9A**, and the second member according to the present invention may be a casing **8A**.

(73) Specifically, the coupler **9A** according to the modification 1-1 includes a coupler main body **95** and an attachment bush **96** as illustrated in FIG. **5**.

(74) The coupler main body **95** has a bottomed-cylindrical shape enabling to be engaged in the eyepiece **232**, similarly to the coupler **9** described in the first embodiment.

(75) In a bottom portion of this coupler main body **95**, a through hole **951** that penetrates through to its front and rear surfaces, and that has a circular shape when viewed from top is formed as illustrated in FIG. **5**. Moreover, on an inner circumferential surface of the through hole **951**, a jut-out portion **952** that extends toward the center axis Ax**2** and that has an annular shape around the center axis Ax**2** as its center is formed on a right end rim side in FIG. **5**. Furthermore, on the inner circumferential surface of the through hole **951**, a screw groove **953** is formed on a left side in FIG. **5** relative to the jut-out portion **952**.

(76) The attachment bush **96** is a member to attach the coupler **9A** to the casing **8A**. This attachment bush **96** includes a bush main body **961** and a jut-out portion **962** as illustrated in FIG. **5**.

(77) The bush main body **961** has an outer diameter substantially the same as an inner diameter of

the through hole **951**, and is formed in a cylindrical shape having an inner diameter larger than the inner diameter of the jut-out portion **952**. Moreover, on an outer circumferential surface of the bush main body **961**, a screw groove **963** is formed as illustrated in FIG. 5.

(78) The jut-out portion **962** protrudes toward the center axis Ax2 from a left end rim side of an inner circumferential surface of the bush main body **961** in FIG. 1, and is formed in an annular shape having an inner diameter substantially the same as the jut-out portion **952**.

(79) The coupler main body **95** and the attachment bush **96** are fixed to each other, by engaging the screw grooves **953** and **963** with each other. In this state, a center axis of the attachment bush **96** agrees with the center axis Ax2 of the coupler main body **95**.

(80) In the state in which the coupler main body **95** and the attachment bush **96** are fixed to each other, with surfaces of respective ends of the jut-out portions **952** and **962** facing each other, and the inner circumferential surface of the bush main body **961**, a concave portion **97** extending in an annular shape having the first axis Ax1 in center is provided as illustrated in FIG. 5. An inner surface of the concave portion **97** (the surfaces of respective ends of the jut-out portions **952** and **962** facing each other, and the inner circumferential surface of the bush main body **961**) corresponds to the first rotation-sliding surface according to the present invention.

(81) Moreover, the casing **8A** according to the modification 1-1 is structured without the attachment bush **82**, omitting from the casing **8** described in the first embodiment above, and is constituted of only of the casing **81A** having a protrusion portion **811A** in a different shape from the protrusion portion **811**.

(82) The protrusion portion **811A** has a bottomed-cylindrical shape that communicates inside and outside of the casing main body **81A**, similarly to the protrusion portion **811** described in the first embodiment. Moreover, in the protrusion portion **811A**, the optical device **83** is fixed as illustrated in FIG. 5. Furthermore, in the protrusion portion **811A**, a convex portion **814** that protrudes out from the left end rim on the outer circumferential surface in FIG. 5, and that has an annular shape having the first axis Ax1 as its center is provided. This convex portion **814** has thickness substantially the same as a width of the concave portion **97**. Moreover, the convex portion **814** has an outer diameter substantially the same as the inner diameter of the bush main body **961**. The coupler **9A** is attached to the casing **8A** as the convex portion **814** engages with (fits in) the concave portion **97**. In this state, the center axis Ax2 agrees with the first axis Ax1. Moreover, the coupler **9A** is restricted its movement to a direction of the first axis Ax1 relative to the casing **8A**, and is rotatable relative to the casing **8A** about the first axis Ax1 (the center axis Ax2).

(83) The outer surface of the convex portion **814** is a portion that slides on the inner surface of the concave portion **97**, and corresponds to the second rotation-sliding surface according to the present invention.

(84) At a distal end of the convex portion **814**, a groove portion **815** is formed as illustrated in FIG. 5 or FIG. 6.

(85) The groove portion **815** extends in a spiral shape about the first axis Ax1 at the distal end of the convex portion **814** as illustrated in FIG. 6. In the state in which the coupler **9A** is attached to the casing **8A**, respective parts of the both end portions **816** and **817** in the direction of the first axis Ax1 in the groove portion **815** are positioned outside the concave portion **97**, respectively.

(86) That is, the outer surface of the convex portion **814** (the second rotation-sliding surface according to the present invention) abuts only on a part of the inner surface of the concave portion **97** (the first rotation-sliding surface according to the present invention) by having the groove portion **815**, in the state in which relative rotation of casing **8** and the coupler **9A** about the first axis Ax1 is stopped, to expose a part of the inner surface of the concave portion **97**.

(87) As the modification 1-1 described above, also when the first member according to the present invention is the coupler **9A**, and the second member according to the present invention is the casing **8A**, effects similar to those of the first embodiment described above are produced.

Second Embodiment

(88) Next, a second embodiment of the present invention will be described.

(89) Hereinafter, same reference symbols are assigned to same components as the first embodiment described above, and detailed description thereof is omitted or simplified. FIG. 7 is a cross-section illustrating a connecting portion between the eyepiece 232 and an endoscope camera head 6B according to the second embodiment. FIG. 8 is a diagram illustrating a coupler 9B according to the second embodiment. Specifically, FIG. 8 is a diagram of a view of the coupler 9B from a side on which the eyepiece 232 is fitted in.

(90) In the endoscope camera head 6B according to the second embodiment, as illustrated in FIG. 7 or FIG. 8, the coupler 9B in a different shape from the coupler 9 is used for the endoscope camera head 6 described in the first embodiment above.

(91) The coupler 9B has a through hole 92B having an inner diameter larger than the inner diameter of the through hole 92 when compared with the coupler 9 described in the first embodiment. The inner diameter of this through hole 92B is set to be larger than the outer diameter of the jut-out portions 812, 822. Moreover, on an inner circumferential surface of the through hole 92B, a projection portion 98 that projects toward the center axis Ax2 from a right end rim side in FIG. 7, and in which a proximal end thereof extends in an arc shape having the center axis Ax2 (the first axis Ax1) as its center is formed. This projection portion 98 has thickness substantially the same as the width of the concave portion 84. In the second embodiment, three pieces of the projection portions 98 are provided, and are arranged to be rotationally symmetric at 120° about the center axis Ax2 (the first axis Ax1). Furthermore, a locus of distal ends of the three projection portion 98 forms a circular shape having a diameter substantially the same as the outer diameter of the bush main body 821. The coupler 9B is attached to the casing 8 as the three projection portions 98 respectively engage with (fit in) the concave portion 84 in a state in which an opening of the bottomed-cylindrical shape is directed to the left side in FIG. 7. In this state, the center axis Ax2 agrees with the first axis Ax1. Moreover, the coupler 9B is restricted its movement to a direction of the first axis Ax1 relative to the casing 8, and is rotatable relative to the casing 8 about the first axis Ax1 (the center axis Ax2).

(92) The outer surface of the projection portion 98 is a portion that slides on the inner surface of the concave portion 84, and corresponds to the second rotation-sliding surface according to the present invention.

(93) That is, the outer surface of the projection portion 98 (the second rotation-sliding surface according to the present invention) abuts only on a part of the inner surface (the first rotation-sliding surface according to the present invention) of the concave portion 84 in a state in which relative rotation of the casing 8 and the coupler 9B about the first axis Ax1 is stopped, to expose a part of the inner surface of the concave portion 84.

(94) According to the second embodiment described above, following effects are produced.

(95) FIG. 9 and FIG. 10 are diagrams illustrating an effect of the second embodiment. Specifically, FIG. 9 is a diagram corresponding to FIG. 7. FIG. 10 is a diagram corresponding to FIG. 8.

(96) The endoscope camera head 6B according to the second embodiment includes the casing 8 having the concave portion 84 that extends in an annular shape having the first axis Ax1 as its center, and the coupler 9B having the projection portion 98, the distal end of which extends in an arc shape having the first axis Ax1 as its center and engages with the concave portion 84. The outer surface of the projection portion 98 abuts only on a part of the inner surface of the concave portion 84 in a state in which relative rotation of the casing 8 and the coupler 9B about the first axis Ax1 is stopped, to expose a part of the inner surface of the concave portion 84 outside.

(97) Therefore, the inner surface of the concave portion 84 can be cleaned as described below.

(98) That is, as illustrated in FIG. 9 or FIG. 10, in a gap between a rim portion of the through hole 92B and the jut-out portion 812, a cleaning brush Br is brought into contact with the inner surface of the concave portion 84 from a portion at which the projection portion 98 is not formed. By rotating the coupler 9B and the cleaning brush Br about the first axis Ax1 with respect to the casing

8 as indicated by an arrow in FIG. **10**, a portion between the inner surface of the concave portion **84** and the outer surface of the projection portion **98** can be cleaned easily and efficiently.

(99) Therefore, according to the endoscope camera head **6B** according to the second embodiment, an effect of reducing cleaning time is produced.

(100) Moreover, in the gap between the rim portion of the through hole **92B** and the jut-out portion **812**, a portion at which the projection portion **98** is not formed is to be a ventilation hole that communicates inside and outside a space Ar. For this, it is possible to suppress formation of condensation on the eyepiece optical system **2321** or the optical device **83**, and to reduce drying time when condensation is formed.

(101) Particularly, three pieces of the projection portions **98** are provided at positions to be rotationally symmetric at 120° about the first axis Ax1. For this, the ventilation holes that communicate inside and outside the space Ar can be spaced uniformly, and it is possible to suppress formation of condensation on the eyepiece optical system **2321** and the optical device **83** effectively, and to significantly reduce drying time when condensation is formed.

Modification 2-1 of Second Embodiment

(102) FIG. **11** and FIG. **12** are diagrams illustrating a modification 2-1 of the second embodiment. Specifically, FIG. **11** is a cross section illustrating a connecting portion between a casing **8C** and the coupler **9A** according to the modification 2-1. FIG. **12** is a view of the casing **8C** from a distal end side of a protrusion portion **811C**.

(103) In the endoscope camera head **6B** according to the second embodiment described above, the first member according to the present invention is the casing **8**, and the second member according to the present invention is the coupler **9B**, but it is not limited thereto. For example, as an endoscope camera head **6C** according to the modification 2-1 illustrated in FIG. **11** and FIG. **12**, the first member according to the present invention may be the coupler **9A**, and the second member according to the present invention is the casing **8C**.

(104) Specifically, the coupler **9A** according to the modification 2-1 has the same configuration as the coupler **9A** described in the modification 1-1.

(105) Moreover, the casing **8C** according to the modification 2-1 is structured without the attachment bush **82**, omitting from the casing **8** described in the second embodiment, and is constituted of only of a casing main body **81C** having the protrusion portion **811C** in a different shape from the protrusion portion **811** as illustrated in FIG. **11** or FIG. **12**.

(106) The protrusion portion **811C** has a cylindrical shape that communicates inside and outside the casing main body **81C**, similarly to the protrusion portion **811** described in the second embodiment. Moreover, inside the protrusion portion **811C**, the optical device **83** is fixed as illustrated in FIG. **11**. Furthermore, in the protrusion portion **811C**, a projection portion **818** that protrudes from a left end rim side in FIG. **11** on an outer surface, and a proximal end of which extends in an arc shape having the first axis Ax1 as its center is formed. This projection portion **818** has thickness substantially the same as the width of the concave portion **97**. In the modification 2-1, three pieces of the projection portions **818** are provided, and are arranged to be rotationally symmetric at 120° about the first axis Ax1. Furthermore, a locus of distal ends of the three projection portions **98** forms a circular shape having a diameter substantially the same as the inner diameter of the bush main body **961**. The coupler **9A** is attached to the casing **8C** as the three projection portions **818** respectively engage with (fit in) the concave portion **97** in a state in which an opening of the bottomed-cylindrical shape is directed to the left side in FIG. **11**. In this state, the center axis Ax2 agrees with the first axis Ax1. Moreover, the coupler **9C** is restricted its movement to a direction of the first axis Ax1 relative to the casing **8C**, and is rotatable relative to the casing **8C** about the first axis Ax1 (the center axis Ax2).

(107) The outer surface of the projection portion **818** is a portion that slides on the inner surface of the concave portion **97**, and corresponds to the second rotation-sliding surface according to the present invention.

(108) That is, the outer surface of the projection portion **818** (the second rotation-sliding surface according to the present invention) abuts only on a part of the inner surface of the concave portion **97** (the first rotation-sliding surface according to the present invention), in the state in which relative rotation of casing **8C** and the coupler **9A** about the first axis **Ax1** is stopped, to expose a part of the inner surface of the concave portion **97**.

(109) As in the modification 2-1 described above, also when the first member according to the present invention is the coupler **9A**, and the second member according to the present invention is the casing **8C**, a similar effect as the second embodiment described above is produced.

Modification 2-2 of Second Embodiment

(110) FIG. **13** is a diagram illustrating a modification 2-2 of the second embodiment. Specifically, FIG. **13** is a diagram corresponding to FIG. **7**.

(111) In the second embodiment described above, a projection portion **98D** may be used instead of the projection portion **98** as in an endoscope camera head **6D** (coupler **9D**) according to the modification 2-2 illustrated in FIG. **13**.

(112) The projection portion **98D** includes a projection-portion main body **981** and a convex portion **982** as illustrated in FIG. **13**.

(113) The projection-portion main body **981** is a portion that protrudes toward the center axis **Ax2** from a right end rim side in FIG. **13** on the inner circumferential surface of the through hole **92B**. A proximal end of this projection-portion main body **981** extends in an arc shape having the center axis **Ax2** (the first axis **Ax1**) as its center.

(114) Moreover, the projection-portion main body **981** has thickness larger than the width of the concave portion **84**. Furthermore, a locus of distal ends of the three projection portions **981** forms a circular shape having a diameter substantially the same as the outer diameter of the jut-out portion **812**.

(115) The convex portion **982** protrudes toward the center axis **Ax2** from a left end rim side in FIG. **13** at the distal end of the projection-portion main body **981**. This convex portion **982** has thickness substantially the same as the width of the concave portion **84**. Moreover, a locus of distal ends of the three convex portions **982** forms a circular shape having a diameter smaller than the outer diameter of the jut-out portion **812**, and larger than the outer diameter of the bush main body **821**.

(116) As described above, the distal end of the projection portion **98D** is formed in a stepped-shape along the center axis **Ax2** by the convex portion **982**.

(117) With the structure described above, the coupler **9D** is attached to the casing **8** as the three convex portions **982** respectively engage with (fit in) the concave portion **84** in a state in which an opening of the bottomed-cylindrical shape is directed to a left side in FIG. **13**, and as the distal end of the projection-portion main body **981** abuts on the outer circumferential surface of the jut-out portion **812**. In this state, the center axis **Ax2** agrees with the first axis **Ax1**. Moreover, the coupler **9D** is restricted its movement to the direction of the first axis **Ax1** relative to the casing **8**, and is rotatable relative to the casing **8** about the first axis **Ax1** (the center axis **Ax2**).

(118) The inner surface of the concave portion **84** and the outer circumferential surface and the jut-out portion **812** are portions on which the coupler **9D** slides, and corresponds to the first rotation-sliding surface according to the present invention. Moreover, the outer surface of the projection portion **98D** (the distal end of the projection-portion main body **981** and the outer surface of the convex portion **982**) is a portion that slides on the inner surface of the concave portion **84** and on the outer circumferential surface of the jut-out portion **812**, and corresponds to the second rotation-sliding surface according to the present invention.

(119) That is, the outer surface of the projection portion **98D** (the second rotation-sliding surface according to the present invention) abuts only on a part of the inner surface of the concave portion **84** and of the outer circumferential surface of the jut-out portion **812** (the first rotation-sliding surface according to the present invention) in a state in which relative rotation of the casing **8** and the coupler **9D** about the first axis **Ax1** is stopped, to expose a part of the inner surface of the

concave portion **84** and of the outer circumferential surface of the jut-out portion **812** outside.

(120) According to the modification 2-2 described above, in addition to the effect similar to the second embodiment described above, a following effect is produced.

(121) In the endoscope camera head **6D** according to the modification 2-2, the outer circumferential surface of the jut-out portion **812** and the distal end of the projection-portion main body **981** form a part of the first and the second rotation-sliding surfaces according to the present invention. That is, not a portion deep inside, but a portion easy to clean forms a part of the first and the second rotation-sliding surfaces according to the present invention. Therefore, cleaning is made further easier, and cleaning time can be further reduced.

Modification 2-3 of Second Embodiment

(122) FIG. **14** is a diagram illustrating a modification 2-3 of the second embodiment. Specifically, FIG. **14** is a diagram corresponding to FIG. **7**.

(123) In the second embodiment described above, as in an endoscope camera head BE (coupler **9E**) according to the modification 2-3 illustrated in FIG. **14**, a projection portion **98E** may be used instead of the projection portion **98**.

(124) The projection portion **98E** is arranged at a position offset to right side (side apart from the eyepiece **232**) from the projection portion **98** described in the second embodiment.

(125) According to the modification 2-3 described above, in addition to an effect similar to the second embodiment, a following effect is produced.

(126) In the endoscope camera head **6E** according to the modification 2-3, the projection portion **98E** is arranged at a position offset to a direction apart from the eyepiece **232**. That is, by expanding the space Ar, sufficient ventilation is achieved. Therefore, it is possible to suppress formation of condensation on the eyepiece optical system **2321** or the optical device **83** effectively, and to significantly reduce drying time when condensation is formed.

Other Embodiments

(127) Embodiments to implement the present invention have been described, but the present invention is not limited only to the first and the second embodiments and the modifications 1-1, 2-1 to 2-3 described above.

(128) FIG. **15** is a diagram illustrating a modification 3-1 of the first and the second embodiments.

(129) In the first and the second embodiments and the modifications 1-1, 2-1 to 2-3, the endoscope camera heads **6**, **6A** to **6E** are detachably provided with respect to the resectoscope **2** for urinary organs, but it is not limited thereto. As in an endoscope device **1F** according to the modification 3-1 illustrated in FIG. **15**, the endoscope camera head **6** may be detachably provided, for example, with respect to an endoscope **2F** for digestive organs.

(130) The endoscope **2F** is constituted of a rigid endoscope. That is, the endoscope **2F** is hard or at least a part thereof is flexible and has a thin and long shape, to be inserted into a living body. In this endoscope **2F**, an optical system that is constituted of one or more lenses, and gathers a subject image is provided. Light supplied from a light source device **14** (FIG. **15**) to the light guide **2322** is emitted from a distal end of the endoscope **2F**, to be irradiated to an inside of the living body. The light irradiated inside the living body and reflected inside the living body (subject image) is gathered by the optical system in the endoscope **2F**. The endoscope camera head **6** images the subject image gathered by the optical system in the endoscope **2F**.

(131) Moreover, not limited to a rigid endoscope, the endoscope **2F** may be a flexible endoscope.

(132) In the first and the second embodiments, and the modifications 1-1, 2-1 to 2-3, 3-1, the endoscope devices **1**, **1F** may be an endoscope device that is used in an industrial field, and that observes inside a subject, such as a mechanical structure.

(133) In the first embodiment and the modification 1-1 described above, the grooves according to the present invention (the groove portions **94**, **815**) are arranged in the convex portions **93**, **814**, but it is not limited thereto, and may be arranged in the concave portions **84**, **97**.

(134) In the second embodiment and the modifications 2-1 to 2-3 described above, the number of

the projection portions according to the present invention (the projection portions **98, 98D, 98E, 818**) is not limited to three, but may be one, two, four, or more.

REFERENCE SIGNS LIST

(135) **1, 1F** ENDOSCOPE DEVICE **2** RESECTSCOPE **2F** ENDOSCOPE **3** ENDOSCOPE IMAGING DEVICE **4** DISPLAY DEVICE **5** CONTROL DEVICE **6, 6A TO 6E** ENDOSCOPE CAMERA HEAD **7** CABLE **8, 8A, 8C** CASING **9, 9A, 9B, 9D** COUPLER **10** PRISM **11** LENS UNIT **12** IMAGING UNIT **13** OPERATING UNIT **14** LIGHT SOURCE DEVICE **21** SHEATH **22** GUIDE TUBE **23** ENDOSCOPE **24** RESECT ELECTRODE MEMBER **25** HANDLE PORTION **81, 81A, 81C** CASING MAIN BODY **82** ATTACHMENT BUSH **83** OPTICAL DEVICE **84** CONCAVE PORTION **91** PRESSURIZING PORTION **92, 92B** THROUGH HOLE **93** CONVEX PORTION **94** GROOVE PORTION **95** COUPLER MAIN BODY **96** ATTACHMENT BUSH **97** CONCAVE PORTION **98, 98D, 98E** PROJECTION PORTION **221** MOUNTING MEMBER **222** WATER SUPPLY INLET **231** INSERTION PORTION **232** EYEPIECE **251** FIXING RING **252** SLIDER **253** SPRING MEMBER **811, 811A, 811C** PROTRUSION PORTION **812** JUT-OUT PORTION **813** SCREW GROOVE **814** CONVEX PORTION **815** GROOVE PORTION **816, 817** BOTH END PORTION **818** PROJECTION PORTION **821** BUSH MAIN BODY **822** JUT-OUT PORTION **823** SCREW GROOVE **941, 942** BOTH END PORTION **951** THROUGH HOLE **952** JUT-OUT PORTION **953** SCREW GROOVE **961** BUSH MAIN BODY **962** JUT-OUT PORTION **963** SCREW GROOVE **981** PROJECTION-PORTION MAIN BODY **982** CONVEX PORTION **2321** EYEPIECE OPTICAL SYSTEM **2322** LIGHT GUIDE **2323** LIGHT SOURCE CONNECTOR **2324** EXPANDING PORTION **2521** HIGH-FREQUENCY POWER CORD **2522** POWER SOURCE CONNECTOR **2523** FINGER HOOK MEMBER **A_r** SPACE **A_{x1}** FIRST AXIS **A_{x2}** CENTER AXIS **A_{x3}** ENDOSCOPE OPTICAL AXIS **A_{x4}** IN-CASE OPTICAL AXIS **CN1, CN2** CONNECTOR **O** CENTER OF GRAVITY

Claims

1. An endoscope camera head comprising a first member and a second member that are capable of rotating relative to each other about a first axis, wherein the first member has a concave portion that extends in an annular shape about the first axis as a center is provided, and a first rotation-sliding surface that extends in an annular shape about the first axis as a center, wherein the first rotation-sliding surface includes an inner surface of the concave portion, the second member is configured to engage with the first rotation-sliding surface and has a convex portion that extends in an annular shape about the first axis as a center and engages with the concave portion, and a second-rotation sliding surface configured to rotation-slide relatively to the first rotation-sliding surface, wherein the second rotation-sliding surface includes an outer surface of the convex portion, and at a distal end of the convex portion, a groove that extends in a spiral shape, so that the second rotation-sliding surface abuts only on a part of the first rotation-sliding surface, to expose a part of the first rotation-sliding surface outside.
2. The endoscope camera head according to claim 1, wherein respective parts of both end portions of the groove in a direction of the first axis are positioned outside the concave portion.
3. The endoscope camera head according to claim 1, comprising: a coupler holding an endoscope configured to capture and output a subject image; and a camera-head that includes a casing supporting the coupler rotatably about the first axis, and an image sensor housed in the casing, the image sensor being configured to image the subject image emitted from the endoscope, wherein the first member is the casing, and the second member is the coupler.
4. The endoscope camera head according to claim 1, comprising: a coupler holding an endoscope configured to capture and output a subject image; and a camera-head that includes a casing supporting the coupler rotatably about the first axis, and an image sensor housed in the casing, the image sensor being configured to image the subject image emitted from the endoscope, wherein the

first member is the coupler, and the second member is the casing.

5. The endoscope camera head according to claim 1, wherein the second member is configured to engage with the first rotation-sliding surface in a state in which movement to a direction of the first axis is restricted relative to the first member.

6. The endoscope camera head according to claim 1, wherein the second member includes a protrusion that extends along a direction of the first axis.

7. The endoscope camera head according to claim 6, wherein the protrusion is a cylinder.

8. The endoscope camera head according to claim 1, wherein the first member includes a main body that extends from the concave portion along a direction of the first axis.

9. The endoscope camera head according to claim 8, wherein the main body is a cylinder.

10. The endoscope camera head according to claim 8, wherein the main body includes a screw groove on an outer surface thereof.
